Piano string excitation V: Spectra for real hammers and strings

Donald E. Hall
Physics Department, California State University, Sacramento, California 95819-2694

Anders Askenfelt
Department of Speech Communication and Music Acoustics, P.O. Box 70014, Royal Institute of Technology (KTH), S-100 44 Stockholm, Sweden

(Received 30 September 1987; accepted for publication 5 January 1988)

New measurements of piano hammer properties are reported, as well as hammer accelerations during impact and resulting string motion on a grand piano. Comparison with extant theory on string–hammer interaction requires discussion of limitations due to: (1) nonlinear mode coupling for finite amplitude, (2) string stiffness and the resulting dispersion, (3) soundboard admittance, (4) finite hammer width, and (5) nonlinearity in the hammer. Items (1) and (4) are shown to be relatively unimportant for measurements made promptly after striking, while (3) is significant mainly for the uppermost treble octave. Item (2) becomes important above roughly 1 kHz for notes at the bass end, 5 kHz in midrange, and 10 kHz at the treble end. Item (5) is the most important limitation, and leads us to consider either using an "effective compliance" to obtain an approximate spectrum or performing numerical integrations to handle the nonlinearity. With these limitations in mind, an examination is made of how well the theory can explain the measured string vibration spectra. The agreement is moderately good in the bass and midrange while, at the treble end, the measured spectra fall off significantly faster than the predictions. Further progress apparently requires methods capable of fully integrating the nonlinear hammer compliance into the theory.

PACS numbers: 43.75.Mn, 43.40.Cw, 43.40.Kd

LIST OF SYMBOLS

A_n	amplitude of string mode n
a	distance of hammer from string end
$\alpha = a/L$	fractional length to striking point
C	hammer compliance
$C_0 = m/16\mu T$	critical compliance
$c = c_T$	speed of transverse waves
c_L	speed of longitudinal waves
d	string (core) diameter
E_0	initial hammer energy
E_n	string mode energy
$F(t)$	force exerted by hammer
L	string length
μ	linear mass density of string
$M = \mu L$	actual string mass
M_Y	mass equivalent of string stiffness
m	hammer mass
p	compliance nonlinearity exponent
$S = \pi d^2/4$	string cross-section area
s	length equivalent to stiffness load
T	string tension
τ	duration of contact
V	initial hammer speed
ω_n	string mode angular frequency
ω_{st}	maximum angular frequency to neglect string stiffness
W	width of string–hammer contact
$\xi(t)$	hammer felt compression
Y	Young's modulus
$Z(\omega)$	string input impedance

INTRODUCTION

As part of a complete explanation of the sound eventually radiated from a piano soundboard, it is necessary to understand the process of string excitation by striking with a hammer. Modern instrumentation makes it possible to obtain extensive and detailed data both on that process and on the subsequent string vibrations. Our goal in this article is to present some new laboratory measurements on a grand piano and to evaluate how well they can be explained with the available theory.

The preceding articles in this series[1–4] have addressed several aspects of the problem of computing piano string motion, and specifically the spectrum of the free motion after contact with the hammer has ended. We are concerned here especially with the validity of treating the hammer as a point mass,[2] or with an added Hookean compliance[3]. In both cases, the string was treated as perfectly flexible with fixed ends. Within these limitations, the idealized linear system is one for which the motions are calculable, and examples have been given to illustrate how the results depend on the relevant parameters. Those parameters are m/M (the ratio of hammer mass to string mass), $\alpha = a/L$ (the fraction of string length at which the hammer strikes), and C/C_0 (the ratio of hammer compliance to the critical compliance $C_0 = m/16\mu T$).

 0001-4966/88/041627-12$00.80 © 1988 Acoustical Society of America

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

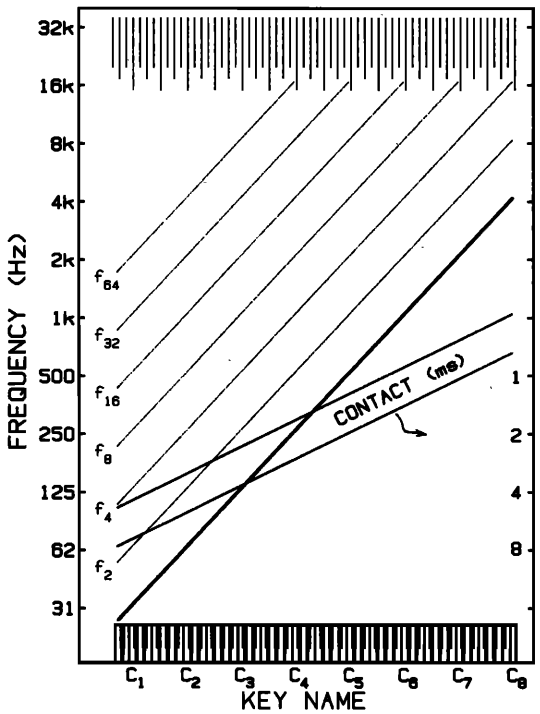


FIG. 1. Frequencies of octave-related string modes as functions of musical scale position, and comparison with typical piano hammer contact times.

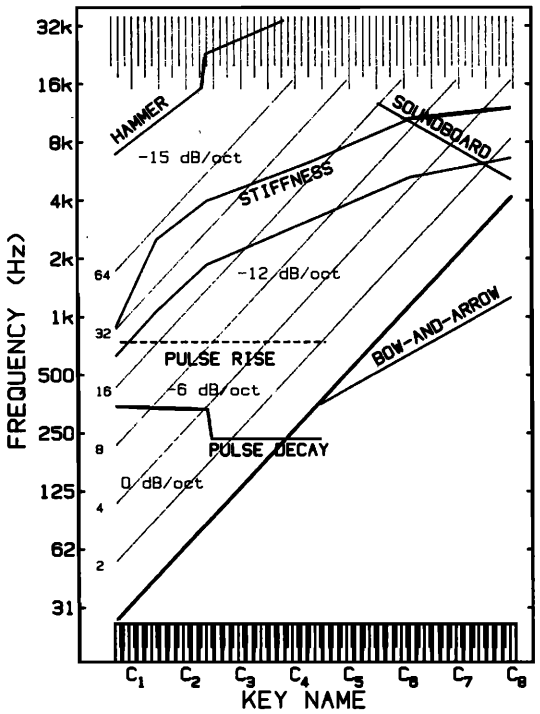


FIG. 2. Approximate frequencies at which various physical effects place limits on validity of theoretical models and expected spectrum slopes. Position of dashed **pulse rise** line depends on hammer compliance. Region of validity of the compliant-hammer model of Ref. 3 lies below the **stiffness** and **soundboard** lines.

In order to make reasonable comparisons of this theory with experiments, we first emphasize the physical factors controlling the overall trends in the mode energy spectra, and then estimate how much the predicted spectra should be modified by each of the factors not yet taken into account. Section III of this article describes various measurements carried out on piano strings and hammers, and Sec. IV compares them with the theoretical predictions.

I. PREDICTED SPECTRAL TRENDS

There are several characteristic time scales associated with the string–hammer system. As a convenient framework for relating such information to the string vibrations over the entire range of a piano keyboard, consider Fig. 1. Here, the number of audible partials for any given note is easily seen, and, when a time scale for any particular physical effect is estimated, it can quickly be seen which parts of the spectrum will be affected. For immediate purposes of this discussion, the slight inharmonicity of mode frequencies and the accompanying tendency toward "stretched tuning" are not included in the figure.

Along with the string mode frequencies, Fig. 1 also shows approximate hammer contact times τ as measured by Quitter.[5] They are plotted to correspond with $f_\tau = 1/2\tau$, the frequency for which one half-cycle has the same duration. This is a reminder that most modes for most strings are excited by a force function $F(t)$ that persists for a relatively long time; only for the lowest mode or two of the extreme bass notes might one make a reasonably good estimate of their excitation by replacing $F(t)$ with one or more delta-function impulses. This extended duration in the applied force was the original motivation for the theory presented in Refs. 1–3. Estimates based on this theory are presented in Fig. 2 to clarify what physical effects are most responsible for the trends in piano string spectra.

A. Treble: Heavy hammers

Consider first the line marked **bow-and-arrow** in Fig. 2. It corresponds to the fundamental frequency of a very light string loaded with a mass m at the striking point:

$$f_1^m \simeq (1/2\pi)\sqrt{T/mL\alpha(1-\alpha)}. \quad (1)$$

In the figure, we have used $\alpha = \frac{1}{8}$ as a typical value for the fractional distance from string end to mass. This model is relevant only for those notes whose hammers are more massive than the strings, which are those in about the upper half of the piano's range. To the extent that the actually sounded frequencies lie well *above* this line, one might hope that the bow-and-arrow model (discussed in Ref. 2, Sec. IV) could indicate how fast the string mode amplitudes A_n or mode energies E_n fall off with n. Its prediction is that $A_n \propto n^{-3} \sin n\pi\alpha$ or $E_n \propto n^{-4} \sin^2 n\pi\alpha$, i.e., falloff at -12 dB/oct beginning already at $n = 1$. In the extreme treble, there is perhaps a large enough mass ratio that we could reasonably look for this dependence in the measurements even though some dissipation must be admitted in the theory to justify it.

B. Bass: Light hammers

The **pulse decay** line in Fig. 2 is determined by how fast the hammer loses its initial velocity into the radiation resis-

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

tance of the string. According to Ref. 2, Eq. (10), the exponential decay rate is about

$$\frac{1}{2\pi}\frac{2T}{mc}=\frac{\mu c}{\pi m}=\frac{Mc}{\pi mL}=\frac{2}{\pi}\frac{M}{m}f_1, \tag{2}$$

and, for moderate compliance, the more complicated expressions given in Ref. 3 do not greatly alter the numerical values. (The step in this line occurs because of the jump in tension at the transition to the bass bridge.) Only for frequencies well *below* this line is the hammer's force concentrated in a sharp enough pulse that the spectrum envelope might resemble the $A_n \propto n^{-1} \sin n\pi\alpha$ or $E_n \propto \sin^2 n\pi\alpha$ of a delta-function impulse. We see that this is to be expected only for the lowest modes in the extreme bass register. Well above this line, we may anticipate a spectrum envelope falling off at least as fast as $A_n \propto n^{-2}$ or $E_n \propto n^{-2}$ (-6 dB/oct) because the force persists through more than a cycle of f_n, as discussed in Ref. 1.

C. Hammer compliance

Insofar as the piano hammer compliance is less than the critical compliance in the linear theory discussed so far (which is generally true except for soft blows at the bass end), the rise time $2TC/c$ of the force pulses felt by the string is much shorter than their decay. Typical values may be as little as 50 μs for hard blows, which would be associated with a further increase of -6 dB/oct in spectrum slope for $f \gg 3$ kHz. The dashed line in Fig. 2 indicates schematically where this might occur for a longer rise time of 200 μs. But remember that this line does not really have a fixed position in the diagram since we will find in Sec. III A that the compliance C is strongly dependent on how vigorously the string is struck. For soft blows in the bass, C may be comparable to C_0; in this case, the rise and fall times are similar, and, instead of two 6-dB/oct breaks, there should be a single transition from flat to -12 dB/oct. These idealized possibilities are illustrated by the solid lines in Fig. 3, all of which correspond to either -6 or -12 dB/oct. The spectra generated by the method of Ref. 3 generally are well fit by such envelopes.

II. LIMITS OF THEORY FOR REAL PIANOS

Let us now examine several effects that might alter the expectations just presented. The remaining lines in Fig. 2, as well as the dashed lines in Fig. 3, will be discussed in Secs. B–D below.

A. Finite amplitude

Only for infinitesimal amplitude do we have a strictly linear differential equation to govern the string vibrations. For finite amplitude, two concerns must be addressed: (1) Is our ability to calculate mode amplitudes from the string–hammer interaction compromised? (2) Is the comparison of predictions with measurements complicated by significant interchange of energy among modes during the subsequent free string motion?

The primary nonlinear effect is the increase in tension when the waves are present, and the consequent raising of natural mode frequencies. For an initial tension T_0, string

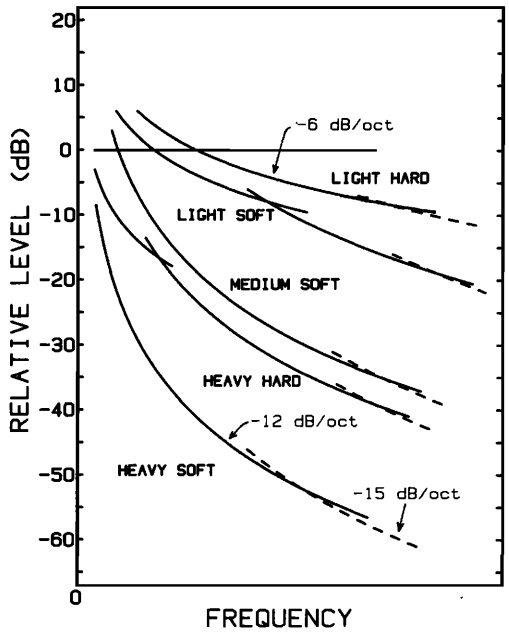


FIG. 3. Approximate upper envelopes that should limit piano string spectra, for hammers with linear compliance. Individual curves may be moved up and down (but not right or left) to fit various cases. Dashed curves represent modification by string stiffness.

cross-sectional area S, and Young's modulus Y, it is easy to show that the mode frequency shifts are on the order of

$$\frac{\Delta\omega_p}{\omega_p}\simeq\frac{YS}{4T_0^2L}\sum E_n<\frac{YS}{4T_0^2L}E_0. \tag{3}$$

Here, each string mode has energy E_n, and the total energy in the string cannot exceed the initial hammer energy $E_0 = mV^2/2$. For the calculations during contact to remain valid, the phase error $\Delta\omega_p\tau$ accumulated during the contact time τ must remain small, so we require

$$(YS/4T_0^2L)E_0\omega_p\tau \ll 1. \tag{4}$$

By using $c_T^2 = T_0/\mu$ and $c_L^2 = YS/\mu$, we can write this in terms of transverse and longitudinal wave speeds as

$$E_0 \ll 4Mc_T^4/c_L^2(\omega_p\tau). \tag{5}$$

In our experiments with hammers described below, the hardest blows we produced had initial hammer speed $V \sim 5$ m/s or energy $E_0 \sim 0.12$ J. We find that such blows still satisfy (5) for nearly all modes in the audible range on middle and high notes, and even in the low bass register for at least the first dozen modes.

Our other concern is whether mode interaction alters the spectrum appreciably during the time of observation of the free string motion. This problem has been addressed by Legge and Fletcher,[6] who argue that the lowest-order nonlinearity is not capable of causing secular energy exchange among the modes unless aided by finite bridge admittance. On a piano, that admittance is small enough that energy will only be exchanged on a long time scale. This is unimportant for present purposes, since we would already want to compare theoretical predictions only with measurements of the early string motion (say within the first few dozen cycles after the hammer blow) in order not to be concerned with the different damping rates of different modes.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

B. String stiffness

Lack of perfect flexibility in the string has two main effects: It becomes impossible for the string to have any discontinuity in slope at the striking point, and the string is now a dispersive medium on which any traveling wave other than a pure sinusoid must change shape as it moves. Strictly speaking, this not only nullifies the theory of Refs. 2 and 3 but also requires that any better theory must take a different approach from the beginning. Insofar as the stiffness is small, however, so also is the dispersion, and the radius of curvature at the struck point can be small enough to resemble a kink if not examined too closely. Therefore, the available predictions should remain accurate as long as we do not consider too small a spatial scale or too high a frequency.

To estimate the frequency at which the stiffness becomes important, think of it as requiring that a small piece of the string be carried along nearly rigidly by the hammer. That is, a small mass load M_Y is encountered in addition to the purely resistive load $2T/c$ of the flexible string, as suggested in Fig. 4. Although calculation of the actual transient dispersive waveform would be very difficult, we can understand what happens qualitatively by thinking about the impedance an infinite string would present to a steady sinusoidal driving force. This is

$$Z(\omega) = (2T/c)[R(\omega) + jX(\omega)], \tag{6}$$

where the normalized resistance R and reactance X are most conveniently written in terms of a dimensionless combination w as

$$R^2 = (1/2)(1 + \sqrt{1 + w^2}), \tag{7}$$

$$X^2 = (1/2)(-1 + \sqrt{1 + w^2}), \tag{8}$$

$$w^2 = (\omega S/c)^2(Y/\pi T). \tag{9}$$

For small stiffness or low frequency,

$$R(\omega) \simeq 1 + w^2/2, \tag{10}$$

$$X(\omega) \simeq w/2 = (\omega S/2c)\sqrt{Y/\pi T}, \tag{11}$$

which says that the effective mass load due to the stiffness is about

$$M_Y \simeq \mu S\sqrt{Y/\pi T}, \tag{12}$$

or that of a short piece of string of length

$$s \simeq S\sqrt{Y/\pi T}. \tag{13}$$

(This is written in such a way as to apply even to a wrapped string, if Y and S apply to the core alone, while μ includes both core and winding.) Values of s range from about 5 mm in the treble up to 10 mm or more in the bass.

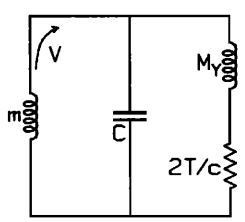


FIG. 4. A circuit analogy for striking an infinite string with a hammer of mass m, compliance C, and initial speed V.

The reactance of the mass load is much less than the resistance if $w \ll 1$, which gives the estimate

$$\omega_{st} \simeq (c/S)\sqrt{\pi T/Y} \tag{14}$$

for the angular frequency above which flexible-string theory is not appropriate. It is this estimate that has been used to plot the upper of the two **stiffness** lines in Fig. 2. Above this line, the spectrum envelope should drop off another 3 dB/oct faster than otherwise predicted, as indicated by the dashed lines in Fig. 3. (This is 3 dB/oct, rather than 6 dB/oct as one might expect, because for $w \gg 1$ the impedance Z grows only in proportion to $\sqrt{\omega}$.)

An alternative estimate can be made by asking that the raising of mode frequencies by the stiffness not destroy the accuracy of the predictions by significantly altering phase relations during the contact time. Since the mode frequencies are given by

$$\omega_n = (n\pi c/L)(1 + n^2\pi^3 Y d^4/128TL^2), \tag{15}$$

the requirement $\Delta\omega_n \tau < 1$ gives

$$n^3 < 128TL^3/\pi^4 cYd^4\tau. \tag{16}$$

This give the lower **stiffness** line in Fig. 2. We see that only the extreme high-frequency end of the spectrum, where amplitudes are already very weak, is affected for most notes. But, for the uppermost octive at the treble end, predictions about modes beyond $n = 2$ or 3 need to take stiffness into account.

C. Soundboard admittance

Yielding of the string supports can, in principle, alter both the amplitude and shape of a reflected wave. These changes are likely to be rather small, however, and may be conveniently imagined to vitiate our predictions in two distinct ways. The first of these is simple reduction in amplitude due to energy loss into the real part of the soundboard admittance. As discussed in Ref. 4, this might be on the order of 10% reduction in the worst cases, which would be at the treble end. These are also worst in the sense that greater numbers of reflections occur before contact ends. The criterion that mode amplitudes not be reduced by more than half during the contact time has been used to estimate the **soundboard** line in Fig. 2.

The second possibility is that the imaginary part of the soundboard admittance has different phases for different frequencies, so that the effective point of reflection differs for different modes. Requiring that information about the relative phases not be entirely lost leads again to roughly the same estimate as in the preceding paragraph. We see that this effect interferes with predictions of initial spectra only in about the last octave at the treble end. Otherwise, it is important only as the mechanism for the gradual decay of mode energies in the free string motion.

D. Finite hammer width

If we could assume that the total force exerted by the hammer is unaltered by giving it finite contact width W, and that the same force density F/W acts throughout that width, the effect of the width would be explicitly calculable. Follow-

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

ing Eq. (A3) of Ref. 1, the effectiveness of a given force for exciting the nth mode is simply reduced by a factor $(\sin z)/z$, where $z = n\pi W/2L$. This represents an additional 6-dB/oct rolloff for mode numbers above $n = 2L/\pi W$, as has been discussed qualitatively by Benade.[7] Using rather generous estimates of W ranging from 12 mm in the bass to 6 mm in the treble, we obtain the **hammer** line in Fig. 2.

In reality, the distribution of force across the contact would be nonuniform, controlled by the interaction of the hammer material with the waves on the string. A correct dynamical theory of this process would be a very interesting and challenging piece of mathematical physics, and has not been done to date. But Eq. (A3) of Ref. 1 still applies, so even an altered force distribution cannot greatly change the estimate above.

We see that essentially throughout the range of interest it is an excellent approximation to treat the hammer as a point contact. The only exception would be modes that would otherwise have extremely small amplitude, which can be "restored" slightly by distributing the force nonuniformly, as discussed in Ref. 4.

E. Hammer nonlinearity

In our measurements of hammer properties to be discussed below, we have found, as have other authors,[8-10] that their compliance is not well represented by Hooke's Law. Strictly speaking, this makes the governing equations nonlinear and disallows the use of the analytic traveling-wave expressions that are at the heart of Ref. 3. The straightforward response to this problem is to abandon that approach and depend instead on numerical integration from the beginning. We believe this could be accomplished, and may be of sufficient interest to explore in the future. Such an approach would be more closely related to some simulations by other authors[8,11-13] than is our present theory.

A possible alternative is to attempt retaining the linear theory as an approximation by using an "effective compliance" C_e. The rationale would be that most of the energy transfer into the string must take place when the compression ξ of the hammer felt is greatest and the force F on the string is near its maximum value. Then perhaps

$$C_e \simeq \xi_{\max}/F_{\max} \tag{17}$$

could serve as an estimate that is nearly true during this part of the process. This will result in some underestimate of the duration of the very small forces that act when $\xi \ll \xi_{\max}$ and so, also, of total contact durations. But we are predicting spectra that cover a very large dynamic range, and we might hope that this approximation would usually not cause errors of more than 2 or 3 dB. If so, this would still leave a useful picture of the spectrum that may be recognizably like the measurements.

Since, for a given hammer, C_e depends on the strength of the blow, one can no longer make a unique prediction for that hammer, but must make a series of predictions. While that requires more calculation, it is exactly what is needed to meet the musically important observational fact that spectra do change markedly as the dynamic level moves from *pianissimo* to *fortissimo*. In the comparisons to be made in the following section, we will always suppose that an "effective compliance" is used in the linear theory.

III. MEASUREMENTS

We have undertaken a variety of measurements bearing on the production of piano sound, of which we report here those having to do specifically with the string–hammer interaction and subsequent string motion. Most of these were first done on a Grotrian–Steinweg 2-m grand piano at the Royal Institute of Technology (KTH). Some supplementary experiments have also been done at California State University, on a 9-ft Steinway model D and on a 7-ft Baldwin model F.

A. Hammer properties

Although the actual hammer motion is a rotation, this is well approximated during the impact as a translational motion of the head perpendicular to the string. The hammer mass is sufficiently concentrated in the head that its center of mass and radius of gyration are nearly coincident and directly in line with the contact point. So a good approximation to the effective mass (say within 5%) is easily determined by turning the hammer upside down and letting its normal contact point rest on a scale, while separate support is provided for the axle at the other end of the shank. A typical set of hammers for a modern grand piano runs smoothly from about $m = 12$ g at the bass end to 6 g at the treble, as illustrated in Fig. 5. The range of string masses M is much greater, so that the ratio m/M reaches about 10 for the highest treble

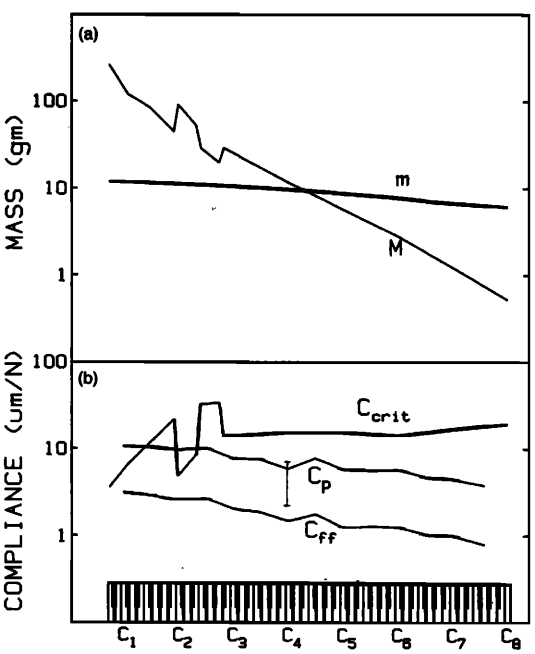


FIG. 5. (a) Comparison of hammer masses m with string masses M in a scaling design by Peter Clark for a 7-ft Steinway model B piano. (b) Comparison of critical compliance $mL/16MT$ with range of effective compliances in a complete hammer set from the Steinweg piano (connected lines) and for one leather-covered hammer (bar). The two outer jumps on the M and C_{crit} curves are the boundaries between single, double, and triple stringing, and the inner jump is the transition from bass bridge to main bridge.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

notes, but may be only about 0.05 in the extreme bass. (Here, it is the total mass of doublets or triplets that is important, not that of a single string.) Neither hammer masses nor treble stringing will differ greatly from one piano to another, but bass string masses might be found to vary with design over a range as great as a factor of 2.

The most straightforward way to measure compliance is to measure the felt compression ξ when it is subjected to a known steady force F, such as a mass load. This was done using loads ranging from 0.55 to 35 N, with resulting deflections up to 1.4 mm. The measurement accuracy was limited to about 0.1 mm. Results for 16 assorted hammers were usually well fit by a power law:

$$F = K\xi^p. \tag{18}$$

Eight sample hammers from an unused set had values of p between 1.5 and 2.8, with no definite trend of p from bass to treble. Of eight other hammers from three different sources, seven had p between 2.2 and 3.5. A Baldwin hammer for C_8 was exceptional, with $p = 5.0$.

We feel it is more important, however, to make a dynamic measurement, since it is questionable whether the behavior of the felt under static compression is the same as during the normal striking process. Therefore, we mounted a piezoelectric force transducer (Bruel & Kjaer 8001) in a drill press chuck with the entire piano action on the platform below it. The hammers could then be thrown against a small aluminum block acting as a face for the transducer by striking the piano keys in the normal manner. (The face of the block was fitted with three pieces of piano wire to simulate the strings, although this made no great difference in the results.) The transducer output was displayed and measured on the screen of a storage oscilloscope, with typical results as in Fig. 6. We tried but abandoned a scheme to make simultaneous optical measurements of hammer body position, as these were complicated by flexing of the hammer shaft and consequent slight rolling motion of the hammer during contact. Thus we have not made the most direct test of the report

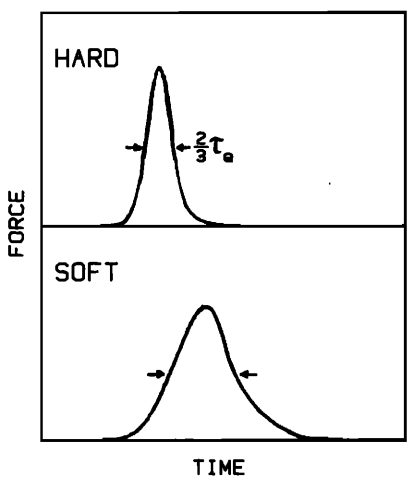


FIG. 6. Oscilloscope traces from force transducer under hard and soft blows from a piano hammer. In top panel, F_{max} is 145 N, width of peak at half-maximum is 0.20 ms, and effective compliance is 1.0 μm/N. Time scale is the same in both cases, but the vertical amplifier is ten times more sensitive for the soft blow; thus $F_{max} = 12$ N, $\frac{2}{3}\tau_c = 0.47$ ms, and $C_e = 2.5$ μm/N.

by Yanagisawa and Nakamura[9] of marked irreversibility in the compression, although it seems to us that the nearly symmetrical shape of our $F(t)$ curves indicates only a very minor role for dissipative forces in the hammer.

Rather than integrating any one of these curves in detail, as Suzuki[10] did, we concentrated on the scaling properties when each hammer was struck with different blows to give peak forces ranging from about 2 to 300 N. One need not find the exact shape of the $F(t)$ curve to notice that $F = ma$ for the force law [Eq. (18)] has self-similar solutions for different amplitudes. By expressing ξ, t, and F as multiples of ξ_{max}, τ, and F_{max}, one may easily show the proportionality,

$$\tau \propto (F_{max})^{(1-p)/2p}, \tag{19}$$

for the contact duration of a blow on a rigid surface. Instead of τ, it was more convenient and accurate to measure the separation between the two times at which F was half of F_{max}, and these times were plotted logarithmically against F_{max}. The results, in most cases, were well approximated by straight lines, whose slopes determine the effective value of the exponent p in Eq. (18).

This procedure was followed with the used hammers from the Steinweg piano, taken at half-octave intervals. The exponents increase smoothly from $p \simeq 2.2$ at C_1 to $p \simeq 3.0$ at C_5, with little further change up to G_7. Values greater than $p = 1$ are not entirely due to peculiarities of the felt; the geometry of the rounded contact would already give $p = 1.5$ (known as Hertz's law) even for locally reacting Hookean material. We also measured a leather-covered midrange hammer of the type used in early pianos, for which $p \simeq 2.3$, although the power-law fit is not as good as for most of the felt hammers. The departure is in the direction of greater nonlinearity for greater forces, suggesting that, beyond a certain point the compliance of the leather is "used up" and that of the harder backing comes into play. We may speculate that those hammer designs that gain the favor of musicians, regardless of superficial differences in materials, will be those that result in $2 < p < 3$. Values too close to $p = 1$ give a nearly linear system that delights the theoretical physicist but has the unmusical property that loud notes are equivalent to amplified soft notes, while $p \gg 3$ probably causes too much contrast between very harsh tone color for *fortissimo* and very bland for *pianissimo*.

Of particular interest is a set of four matched hammers whose compliance was adjusted with "pricking" by a piano technician to represent discrete steps in voicing from "definitely too hard" (H1) through "hard but usable" (H2) and "soft but usable" (H3) to "definitely too soft" (H4). Later, the last of these was pricked still more until obviously "ruined" as far as the technician was concerned (H5). Data on these hammers suggest a progression from $p = 2.3$ for the hardest to 2.8 for the softest and 3.3 after "ruining."

If the hammer were described by Hooke's law ($p = 1$), the $F(t)$ curves in Fig. 6 would be half-cycles of a sine wave, without the extended wings on either side for small values of F. Let us take the "effective compliance" in a given blow to be that constant value C_e which would produce a good match to the upper part of the $F(t)$ curve. Since the half-maximum occurs at 30° and 150° along a sine curve, we iden-

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

tify the width at half-maximum with $(\frac{2}{3})\tau_e$, where the fictional contact time τ_e for $p=1$ would be $\tau_e = \pi/\omega$, with $\omega^2 = 1/mC_e$. Thus, for any blow, we can estimate

$$C_e = (\tau_e/\pi)^2/m, \tag{20}$$

and this varies with the peak force as

$$C_e \propto (F_{\max})^{(1-p)/p}. \tag{21}$$

Then, if extreme values of $F_{\max}$ during playing cover a range as large as 2 to 200 N, the effective compliance of a given hammer could be decreased by as much as a factor of 10 (for $p=2.0$) or 20 (for $p=2.9$) in going from *pianissimo* to *fortissimo*. This is very important in accounting for the differences in sound spectra with dynamic level that are observed. Values of C_e determined in this way fall roughly in the range around 1 to 10 μN/m, as indicated in Fig. 5. The two curves shown are values of effective compliance C_e when the maximum force is 10 N (*p*) or 100 N (*ff*), which we estimate to be typical of the forces reached at those dynamic levels. For the set of four hammers treated by pricking, each successive stage of softening increased C_e by roughly 10% at intermediate force levels, and the final "ruining" of the last hammer added another 20% or 30%. Thus voicing by an exacting piano technician involves adjusting C_e within 5% or 10% of a desired optimum value.

B. String motion

As in earlier work with Jansson,[14] we measured string motion on the Steinweg piano by placing a small horseshoe magnet near one end of a string. Small disk ceramic magnets (thickness 2 mm, diameter 7 mm, field about 0.1 T) served as pole pieces that could fit on opposite sides of the string (or triplet). Since the magnetic field is spread over a finite region (Fig. 7), what is measured is the average velocity over a segment of string not more than about 10 mm long; this is short enough to serve as a point measurement for all the frequencies with which we are concerned. Observed peak velocities ranged from about 0.5 to 6 m/s. The induced voltage between string ends was captured on a storage oscilloscope for waveform documentation, as well as on a digital tape recorder for later spectral analysis. Compared to the optical detection of string position used in Ref. 4, velocity detection has an advantage in being more sensitive to high-frequency modes.

String motion was measured for C_2, C_4, and C_6, each being struck by its own normal hammer (masses 9.8, 8.9, and 6.9 g) and again by an aluminum hammer (8.9 g). In addition, C_4 was struck by each of the set of four hammers of graded compliance, as well as by the leather-covered hammer. For each hammer, motion was recorded for what we judged to be *pp*, *mp*, *mf*, *f*, and *ff* dynamic levels. Reproducible blows for each of these levels were provided by striking the key with an adjustable pendulum. Each of these dynamic steps represented a little less than a doubling of the impulse delivered to the key, so that the total range tested was about 20 to 25 dB.

Typical measured velocities when the detector was located at the midpoint of the short end of the string are illustrated in Fig. 8. In agreement with expectation, C_2 shows a rather narrow pulse from the original impact, followed by its inverted reflection from the agraffe and a second smaller pulse from the hammer. Here, the contact time is less than one-fourth of the string fundamental period, so this part of the string becomes quiescent again until reflected waves begin arriving from the bridge. Slight changes in shape sometimes visible in these reflections represent dispersion due to string stiffness. C_4 shows four distinct reflections on the short end before contact ceases. C_6 has m/M large enough to maintain contact for about one fundamental period. Although, for the felt hammer, there appears to be a single broad pulse, with the aluminum hammer we could see that the motion included about a dozen rapid oscillations representing individual reflected waves. For each note, an increase in angularity and detail of the waveforms is easily visible with increasing dynamic level.

Spectra were computed from samples beginning around a cycle or two after hammer contact ended. Sample lengths were 64, 26, and 18 ms for C_2, C_4, and C_6, respectively.

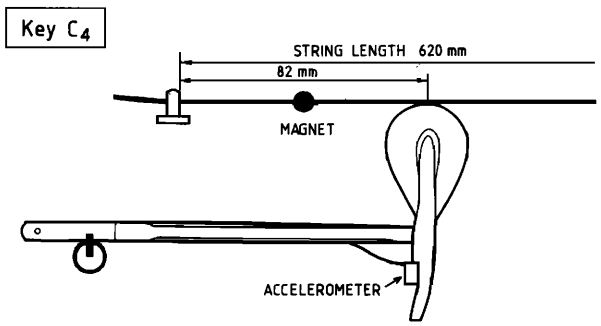


FIG. 7. Relation of hammer, accelerometer, string, and magnetic field region for string velocity measurements; agraffe at left, bridge at right.

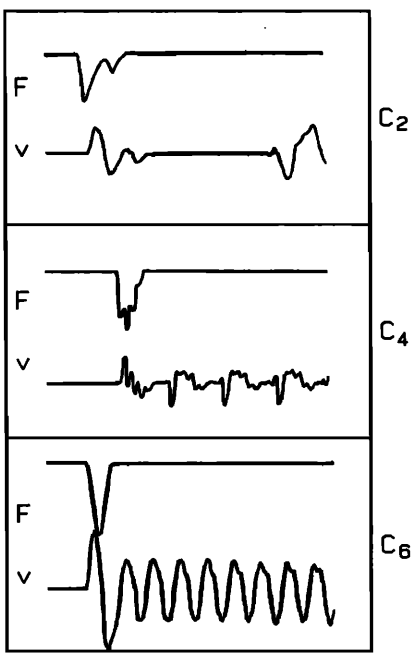


FIG. 8. Sample oscilloscope traces from accelerometer (giving downward force F on hammer) and induced voltage in string (giving velocity v at magnet location) for three notes, all played at *forte* level. Total time shown is 20 ms for C_2 and C_4, 10 ms for C_6.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

Reproducibility was satisfactory both for data samples beginning at various times up to 100 ms or so later and for analyses of repeated blows under the same conditions. For a detector located a distance x from the string end, spectra of the observed signals must be corrected for a factor $\sin n\pi x/L$ representing relative amplitude of mode n at that point. After this correction, the spectra may be equally well regarded as representing either velocity amplitude or mode energy since $E_n \propto v_n^2$. Representative corrected spectra will be shown and discussed in Sec. IV. Those spectra will have gaps around $n = 16$ because the detector was located so close to a node that the correction factor just mentioned is poorly known and prevents reliable determination of the amplitude. Typical uncertainties in the spectra range from as little as 1 dB for the strongest modes up to 3 or 4 dB for the very weak ones.

C. Hammer acceleration

As shown in Fig. 7, a miniature accelerometer (Bruel & Kjaer 4374) was sometimes fastened to the bottom of the hammer head. We found this position better than anywhere on the shank, where extraneous bending vibrations were also detected. Insofar as the hammer constitutes a single lumped mass (an approximation whose limitations have been examined by Suzuki[10]), the accelerometer signal is directly proportional to the force of interaction between hammer and string. This accelerometer has a mass of 0.65 g, so adds between 5% and 10% to the hammer's normal mass. The observed accelerations ranged from about 0.2 to 20 km/s^2, corresponding to forces of roughly 2 to 200 N.

Sample traces from the accelerometer have been shown in Fig. 8. It is perhaps even easier to see here than in the velocity signals the main qualitative features of the interaction: distinctly separated pulses in the bass, several largely overlapping pulses in the midrange, and a large number of pulses merging into a fairly smooth total force in the treble. Again, the pulses from individual reflected waves are more distinct at higher dynamic levels where the hammer has greater effective stiffness, and especially so for the aluminum hammer. It is usually quite easy to tell when contact ends from the accelerometer trace. It is also clear in some of the C_2 samples, especially with the aluminum hammer, that brief loss of contact occurs, followed by recontact upon return of a reflected wave. We also verified these recontacts from watching the resistance changes in a circuit through the piano string and a very fine wire on top of the hammer. (Predictions of recontact have been discussed in Refs. 2 and 3.)

The force pulse does not repeat periodically, so has a continuous Fourier spectrum (Fig. 9). These spectra typically show broad peaks at multiples of the frequency at which waves reflect back and forth on the short end of the string, in keeping with Benade's argument.[15] Following Ref. 1, Eq. (A10), one should be able to obtain the string mode amplitudes by reading values from these force spectra at each mode frequency and correcting for the factor $\sin n\pi\alpha$. (Here, the correction factor lowers the measured levels, whereas the measured string motion levels had to be raised to correct for $\sin n\pi x/L$.) Thus, along with the direct measurement of string velocity, we also have an independent measurement that serves as an indirect check on its accuracy. Figure 10 shows such a comparison, which is fairly successful insofar as each is judged to have possible errors of 2 or 3 dB. (The gaps in the accelerometer spectrum come where near-zero values of $\sin n\pi\alpha$ prevent accurate determination.) Since the accelerometer signal is only an indirect measure of what the string does, complicated by the question of whether the hammer is adequately modeled as a lumped mass, we will use only the directly measured velocity spectra in the comparisons below.

IV. COMPARISON AND CONCLUSIONS

There are two ways in which the success of the present theory should be judged, the first being whether it seems to account correctly for a reasonable amount of detail in indi-

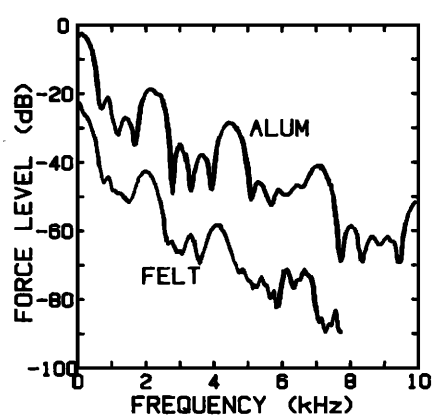


FIG. 9. Sample spectra of force applied to C_4 string, obtained from accelerometer on hammer, for aluminum and felt hammers, at *forte* dynamic level. Reference level is arbitrary, and traces are offset 20 dB for visibility.

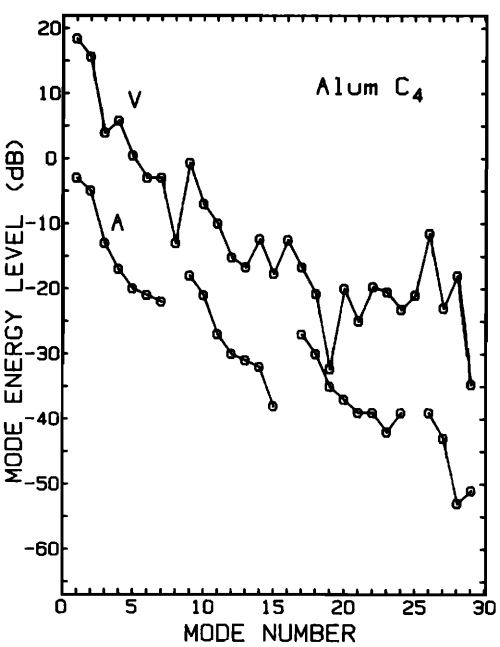
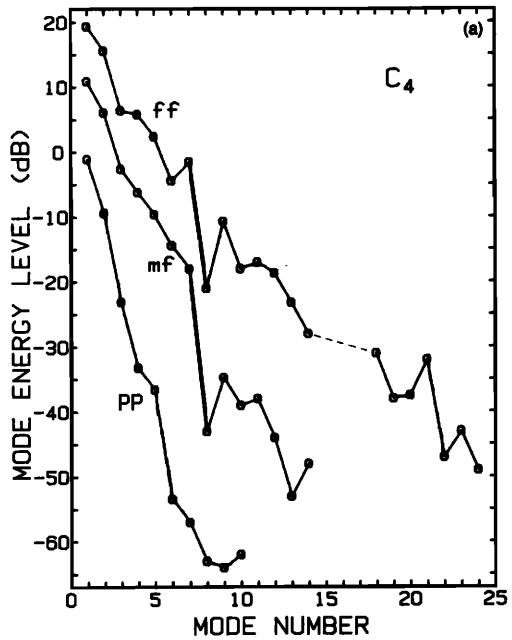
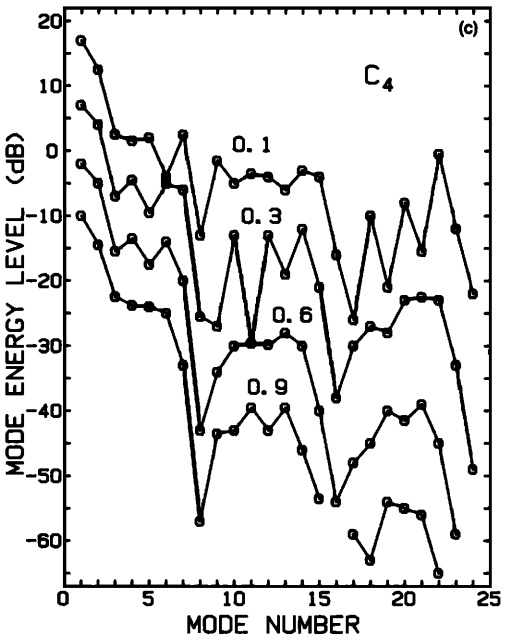


FIG. 10. Comparison of string mode spectra measured directly (V) and computed indirectly from accelerometer signal (A). Offset arbitrarily for visibility, since neither level is absolutely calibrated.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms





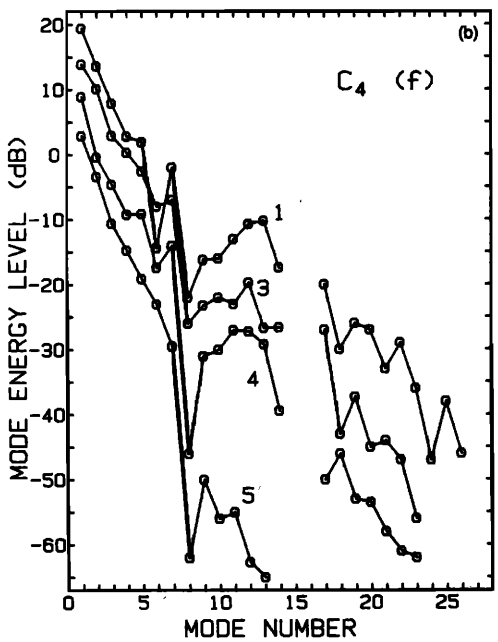


FIG. 11. Comparison of measured and predicted spectra for C_4 ($f = 262$ Hz). Reference level in all measured spectra is arbitrary. (a) String velocity measurements at different dynamic levels for a felt hammer. (b) Measurements at constant dynamic level for hammers treated with varying degrees of pricking, offset in 5-dB steps for visibility. (1 = **too hard**, 3 = **Soft but usable**, 4 = **too soft**, 5 = **ruined**.) (c) Spectra predicted by linear theory, using $m/M = 0.75$ and $\alpha = 0.127$. The parameter is the ratio of effective compliance to the critical compliance of about 14 μm/N.

vidual test cases. We should also examine whether enough insight has been provided to show how broad general features in the spectra depend on the parameters, without the necessity of creating a deluge of numerical calculations of individual cases.

For the first test, consider the spectra for C_4 in Fig. 11(a). For this note, the mass ratio is $m/M = (8.9\text{ g})/(11.8\text{ g}) = 0.75$ and the fractional striking point parameter is $\alpha = (82\text{ mm})/(620\text{ mm}) = 0.132$. Predicted spectra for this value of α, however, lack the observed dip at $n = 8$, with a much better fit being obtained by using $\alpha \simeq 0.127$. The difference agrees well with the inward shift of nodes estimated in the Appendix of Ref. 4 for clamped ends of a stiff string. Similarly, for C_2, good fits were obtained with $\alpha \simeq 0.124$ instead of $(157\text{ mm})/(1239\text{ mm}) = 0.127$. At C_6, the measured spectra did not include enough modes to show clearly whether reducing α from $(16\text{ mm})/(184\text{ mm}) = 0.087$ toward 0.073 was any improvement.

Sample predicted spectra over a reasonable range of compliances are shown in Fig. 11(c). These spectra change only slowly with m/M, so the uncertainty in the value of that parameter is relatively unimportant, but they are quite sensitive to striking point, the resemblance to measurements being distinctly poorer if α changes by as much as 0.002. Detailed agreement should not be sought past roughly the first dozen modes, since Fig. 2 indicates that string stiffness would begin limiting their strength at higher frequencies. It appears that some of the main features in Fig. 11(a) and (c) do resemble each other. But the overall rate of decrease of predicted amplitudes toward higher modes is not as large as

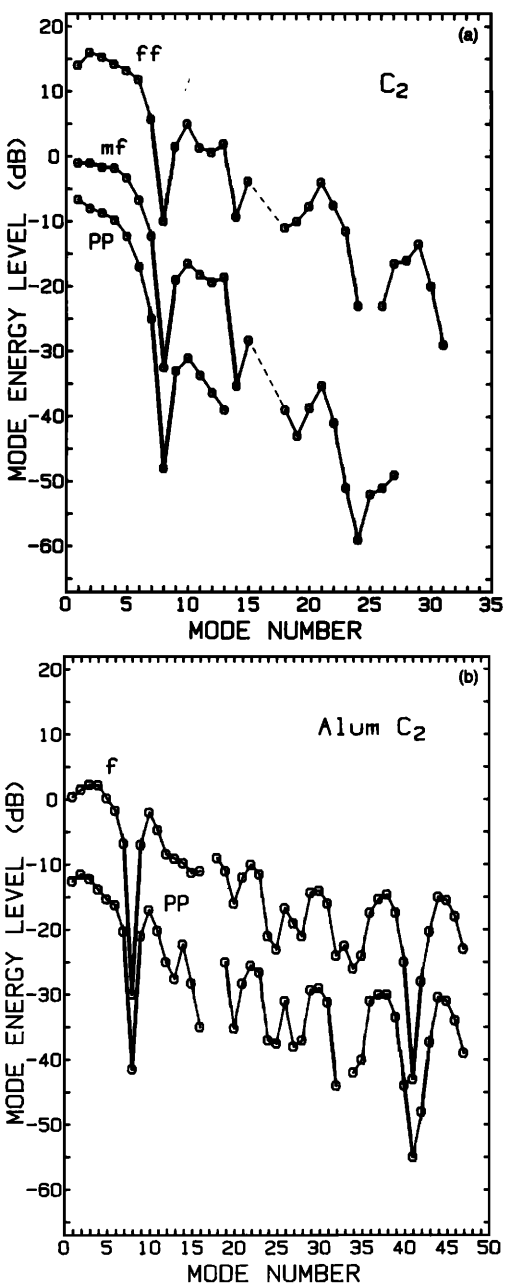
Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

in the observations, especially for the *pp* dynamic level. The same predictions should also match the spectra taken for differently voiced hammers on the same note, shown in Fig. 11(b). A similar fair resemblance is obtained. It is interesting that the "normal" used hammer on this piano seems to be intermediate between cases 4 and 5, indicating that it is too soft or dull according to the standards of the technician who treated the new test hammers.

A similar comparison for C_2 is provided by Fig. 12. In this case, Fig. 2 indicates that agreement might reasonably be sought up to about $n = 25$ or 30. Though the minima at $n = 5$ and 12 are rather washed out, the observed spectra indeed show many of the features predicted, including the dip depth on the order of 20 dB for the nearly missing eighth mode. The aluminum hammer used for Fig. 12(b) may be regarded as providing an extreme version of the "honky-tonk" sound sometimes obtained by putting thumbtacks in piano hammers. Notice that the spectra for aluminum lack the qualitative change with dynamic level seen for the felt hammers; this is because the compliance is so small that its effects (including nonlinearity) simply do not come into play until frequencies higher than those shown.

Finally, we may compare the measurements and predictions for C_6 in Fig. 13. These last predictions were generated only with difficulty, because the mass ratio $m/M = (6.9\ \text{g})/(2.8\ \text{g}) = 2.5$ pushes the limits of the present implementation of the computer program. (Extra tricks were necessary to deal with reflected waves of tenth and higher order without being overwhelmed by roundoff errors.) The agreement for C_6 can hardly be called other than poor.

One consistent feature is that the measured spectra tend to be a bit smoother then the predictions. This is not unreasonable in that the hammer nonlinearity omitted from the theory might be expected to provide a mechanism for mode



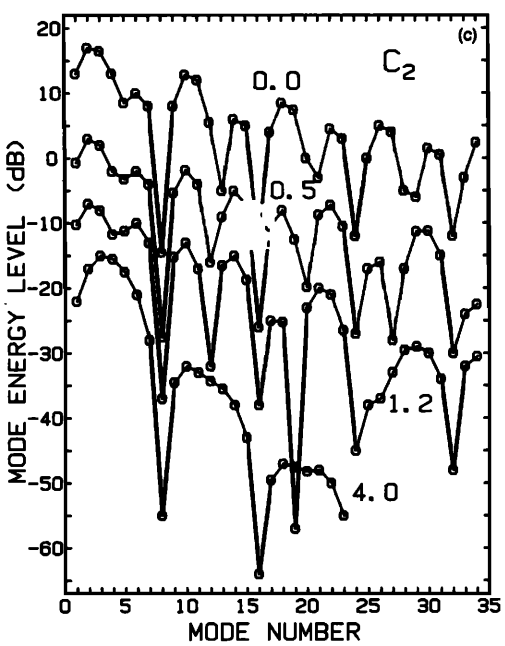


FIG. 12. Comparison of spectra for C_2 ($f = 65$ Hz). (a), (b) Measured spectra at different dynamic levels for a felt and an aluminum hammer. (c) Predictions from linear theory, using $m/M = 0.14$ and $\alpha = 0.124$. The parameter is the ratio of effective compliance to the critical compliance of about 5 μm/N.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

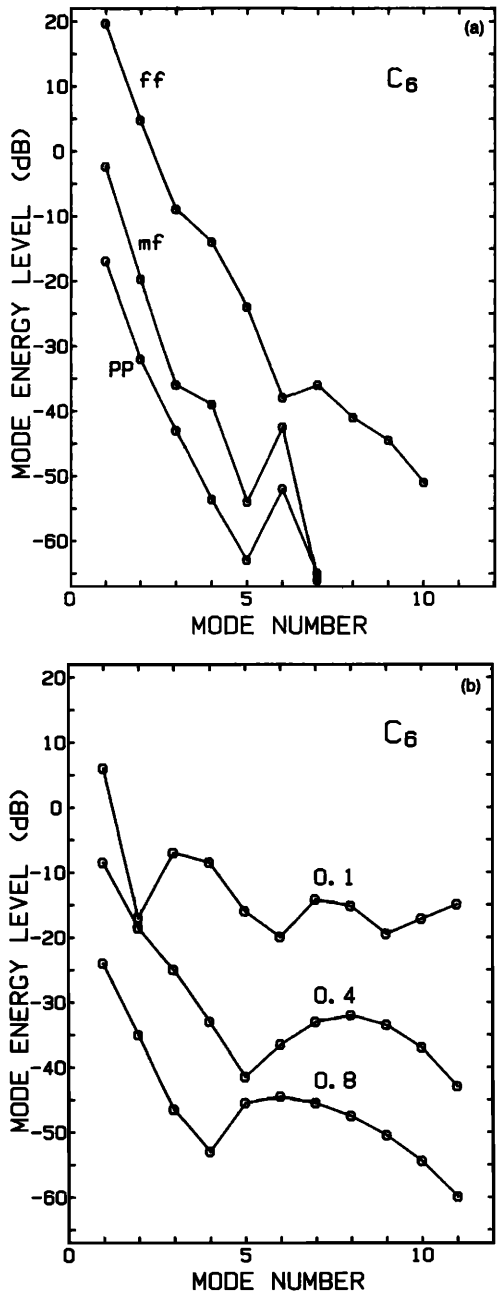


FIG. 13. Comparison of spectra for C_6 ($f = 1047$ Hz). (a) Measured spectra for a felt hammer. (b) Predictions from linear theory, using $m/M = 2.45$ and $\alpha = 0.073$. The parameter is the ratio of effective compliance to the critical compliance of about 11 μm/N.

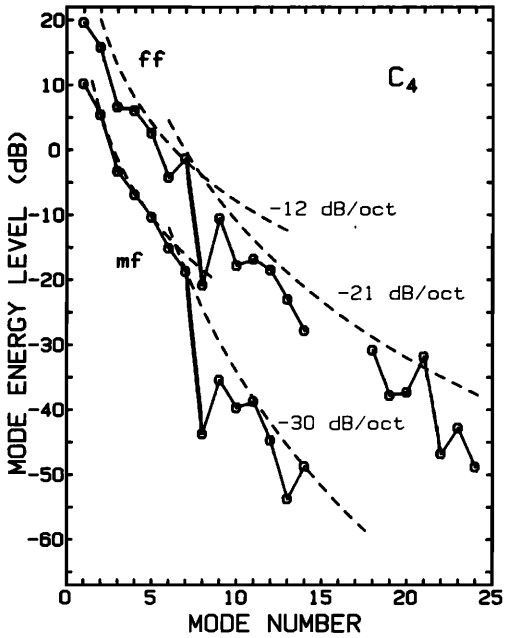


FIG. 14. Observed spectra for C_4 at two dynamic levels, compared with -12-dB/oct envelopes for low frequencies and -21 or -30 dB/oct for high frequencies.

energy exchange during hammer contact. This would more often than not result in transfer of energy from modes that happen to have larger amplitudes in a given case to others that are deficient. (This is not, however, a way of restoring the "missing modes" discussed in Ref. 4.)

Let us examine one case more carefully in order to focus on an important conclusion. Fig. 14 overlays -12-dB/oct envelopes on two sets of data from Fig. 11(a), showing a reasonable resemblance up to $n = 8$ or 10 at most. The expected steepening to -15 dB/oct beyond that point due to string stiffness is definitely not enough to explain the observations. Still faster declines are required in order to explain the data for *pp* blows. This behavior forces us to reexamine the validity of our use of an "effective compliance," and recognize that the original argument for introducing it applies mainly to the first few modes whose periods are longer or comparable to the time scale on which the force $F(t)$ fluctuates. For sufficiently large n, we should argue instead that successive half-cycles of $\exp(j\omega_n t)$ produce very near cancellation when multiplied by the relatively slowly varying $F(t)$ in Eq. (A10) of Ref. 1 for the mode displacement amplitudes:

$$A_n \propto \int_0^\tau F(t)\exp(j\omega_n t)\,dt. \tag{22}$$

In fact, if the most severe singularity anywhere in $F(t)$ is a jump in its qth derivative, then the asymptotic behavior of A_n for large n must be

$$A_n \propto n^{-(q+1)}, \tag{23}$$

which corresponds to $-6(q+1)$ dB/oct. But, for the nonlinear compliance described by (18), $F(t)$ starts out proportional to t^p rather than linearly, so q should be identified here with p. Values of q in the range from 2 to 3 should then lead to spectrum slopes around -18 to -24 dB/oct rather than -12 dB/oct. This argument seems to justify reasonable fits to the *ff* spectra of all three strings, but the steeper slopes for softer blows are still not explained. Further discussion of that problem apparently needs to focus on whether even Eq. (18) is an adequate description of how the contacts are initiated and ended.

We summarize this work in the following conclusions.

(1) For several test cases, we have made measurements of piano string motion and presented them in the form of frequency spectra. Measurements of hammer acceleration provided independent data supporting the validity of these spectra.

(2) The measured spectra have fairly smooth envelopes,

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms

which are usefully described in terms of several possible slopes in decibels per octave corresponding to different dominant physical mechanisms. This is probably the form in which information on string motion would ultimately be most useful as input to the more difficult problem of the soundboard transfer function.

(3) Supplemental measurements of hammer properties show that they depart significantly from Hooke's law compliance.

(4) We have compared these measurements with predictions made with the methods of Ref. 3. In view of the idealizations in the theory and the imperfections of the measurements, there is reasonable agreement at fortissimo levels, but not as good for softer blows. This claim specifically requires us to recognize that most of the spectra extend to high enough frequencies to transgress the limits of validity of the theory. These limits are more severe for treble than for bass notes.

(5) One of the differences is that the measured spectra are somewhat smoother than the predictions, and this may be explained as a consequence of hammer nonlinearity.

(6) The greatest weakness in the theory of Ref. 3 is its use of linear hammer compliance. The theory can still be used in a limited way through the ruse of fictitious "effective compliances" that decrease for a given hammer as the dynamic level of playing increases.

(7) The second greatest limitation of our theory is its use of zero string stiffness, although it is possible to argue that the primary effect of stiffness may only be a relatively minor steepening of spectra at high frequencies.

(8) Finite soundboard compliance is an important limit on the predictions in the uppermost treble octave, but finite hammer size is not important.

(9) Directions that might most profit from further work would include: (a) more extensive work on hammers to determine whether $F \propto \xi^p$ adequately represents their compliance and whether systematic variations of p with size, wear, or manufacturer can be established; (b) measurements of string spectra for a more complete sample of notes (such as one or two per octave all the way from C_1 to C_8), and on pianos of different type and size, with interpretation in terms of spectral slopes; and (c) alteration of the computer programs to include hammer nonlinearity, in hopes of achieving more detailed agreement of the predictions with measurements.

ACKNOWLEDGMENTS

We are indebted to Hans Norén and Conny Carlsson of the Swedish Radio Company and to Peter Clark of California State University for sharing their extensive knowledge and expertise with real pianos. The static compliance measurements were made by Kurt Ryman, a student at CSUS. We also thank the late Professor Arthur Benade and Professor Gabriel Weinreich for stimulating discussions on this subject. The first author is also grateful to Professor Gunnar Fant for the hospitality of his department at KTH during part of this work.

[1]D. E. Hall, J. Acoust. Soc. Am. **79**, 141–147 (1986).
[2]D. E. Hall, J. Acoust. Soc. Am. **81**, 535–546 (1987).
[3]D. E. Hall, J. Acoust. Soc. Am. **81**, 547–555 (1987). Note the following typographical errors in this reference: V in Eq. (22) should be outside the parentheses, and the first minus sign should be plus in the expression for P_{63} in the Appendix.
[4]D. E. Hall and P. Clark, J. Acoust. Soc. Am. **82**, 1913–1918 (1987).
[5]J. P. Quitter, IRE Trans. Audio **6**, 96–103 (1958).
[6]K. A. Legge and N. H. Fletcher, J. Acoust. Soc. Am. **76**, 5–12 (1984).
[7]A. H. Benade, *Fundamentals of Musical Acoustics* (Oxford U.P., New York, 1976), pp. 114–116.
[8]X. Boutillon, "*Le fonctionnement des instruments à cordes libres*," unpublished dissertation, Orsay (1984).
[9]T. Yanagisawa and K. Nakamura, J. Acoust. Soc. Japan **40**, 725–729 (1984).
[10]H. Suzuki, J. Acoust. Soc. Am. Suppl. 1 **78**, S33 (1985).
[11]T. Yanagisawa, K. Nakamura, and I. Shirayanagi, J. Acoust. Soc. Japan **31**, 661–666 (**1975**); also **33**, 412–416 (1977).
[12]R. A. Bacon and J. M. Bowsher, Acustica **41**, 21–27 (1978).
[13]I. Nakamura, *Proceedings of the Wollongong Workshop on the Acoustics of Stringed Musical Instruments* (Catgut Acoustical Society, Montclair, NJ, 1982), pp. 145–177.
[14]A. Askenfelt and E. Jansson, *Proc. Stockholm Music Acoustics Conf.* (Royal Swedish Acad. of Music, Stockholm, 1985), Vol. II, pp. 137–144.
[15]Reference 7, pp. 342–343.

Downloaded 26 Feb 2013 to 134.100.29.35. Redistribution subject to ASA license or copyright; see http://asadl.org/terms